

Grade 6 Accelerated
Unit 11
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

1. Marie's office has a parking garage that charges \$10.50 each day for parking. Marie's employer provides her with a parking card that has \$200 each month. When Marie parked today, the card indicated it had \$116 left on the card. How many days has Marie used the card for parking this month?

$$\frac{(200-116)}{10.50} = 8 \text{ days}$$

2. Tanner is traveling on his first flight to visit family. $8\frac{4}{5}$ minutes into the flight, the pilot announces the plane is currently at an altitude of 3740 feet. At that rate, how long will it take to reach a cruising altitude of 32,400 feet?

$$\frac{8.8}{3740} = \frac{x}{32400} \quad \frac{285,120}{3,740} \approx 76.2 \text{ minutes}$$

3. A penguin sitting on an iceberg notices a fish swimming below. The fish is 10.3 ft below sea level. The penguin dives and reaches the fish in $\frac{7}{15}$ of a minute. What was the rate of descent the penguin made?

$$\frac{10.3 \text{ ft}}{\frac{7}{15} \text{ min}} \approx 22.07 \text{ feet per minute}$$

4. The temperature in North Dakota today was -8.8°C . Over the next 3 weeks, the temperature will increase an average of 3.2°C each week. What will the temperature be at the end of the 3 weeks?

$$3.2(3) = 9.6 \quad -8.8 + 9.6 = 0.8^{\circ}\text{C}$$

5. Lucy buys her cats a bag of food that is 35.3 ounces. Each of Lucy's cats eat $\frac{9}{15}$ ounces a day. If she has 2 cats, how long will the bag of cat food last Lucy?

$$\frac{35.3}{\left(2 \times \frac{9}{15}\right)} = 29\frac{5}{12} \text{ days or approximately 29 days}$$

6. Priscilla is planning to hike a new trail this weekend. She sets up camp Friday evening at the start of the trail which begins at an elevation of -184.56 ft and ends at an elevation of 759.22 ft. She plans to begin her hike in the early morning and wants to travel $\frac{5}{8}$ of the change in elevation of her trip on Saturday, then set up camp until the next morning. What is Priscilla's starting elevation on Sunday?

$$759.22 - (-184.56) = 943.78$$

$$943.78 \cdot \frac{5}{8} = 589.86$$

$$\begin{array}{r} -184.56 \\ + 589.86 \\ \hline 405.3 \text{ ft.} \end{array}$$

7. Kelly is running a race on Monday, which has a total distance of 26.2 miles. Kelly's family wants to show their support by holding up encouraging signs along the path. Her sister decides to stand at the spot that is $\frac{3}{4}$ of the total race while her father chooses to stand at the spot that is $\frac{7}{8}$ of the total race. How many miles apart are Kelly's family members?

$$\begin{array}{ll} \text{Sister} & \text{Father} \\ 26.2 \cdot \frac{3}{4} = 19.65 & 26.2 \cdot \frac{7}{8} = 22.925 \end{array}$$

$$\begin{array}{r} 22.925 \\ - 19.650 \\ \hline 3.275 \text{ mi.} \end{array}$$

8. Kelly runs at a steady rate of 11.2 minutes per mile. Using the information given in Question 7, how many minutes after the race starts should her father expect to see Kelly run by?

$$22.925 \cdot 11.2 = 256.76 \text{ min.}$$

9. Floyd used his store credit card to purchase a new baseball. He has a coupon that when he uses his credit card, the item will be $\frac{2}{5}$ of the total. The baseball was priced at $\$7.25$. A month goes by and Floyd forgot to pay his credit card bill, so they have added a fee of 0.22 times the amount he owed. What is the total amount Floyd has to pay his credit card company?

$$7.25 \cdot \frac{2}{5} = \frac{14.5}{5} = 2.90$$

$$2.90 + 2.90(0.22) = 2.90 + 0.64 = \$3.54$$

10. Stephanie is moving to Miami, Florida from Atlanta, Georgia. The total distance of her move is 665.71 miles. She rented a moving truck that charges $\$22.75$ a day and $\$0.27$ per mile over 500 miles. If she returns the truck with a full tank of gas after 2 days, the company will discount her bill by $\frac{4}{25}$. What is the total amount Stephanie will pay for the moving truck if she fills the tank with gas before returning it after 2 days?

$$\begin{array}{r} 665.71 \\ - 500.00 \\ \hline 165.71 \cdot 0.27 = 44.74 \end{array}$$

$$\begin{array}{r} 22.75(2) = 45.50 \\ + 44.74 \\ \hline 90.24 \cdot \frac{4}{25} = 14.44 \end{array}$$

$$\begin{array}{r} 90.24 \\ - 14.44 \\ \hline \$75.80 \end{array}$$